

CIVIL WORKS FOR SHP

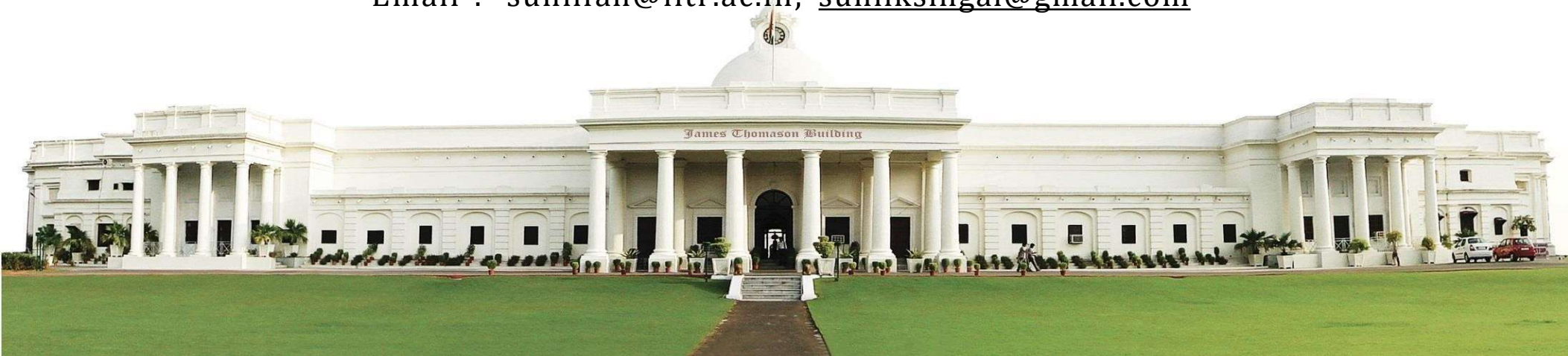
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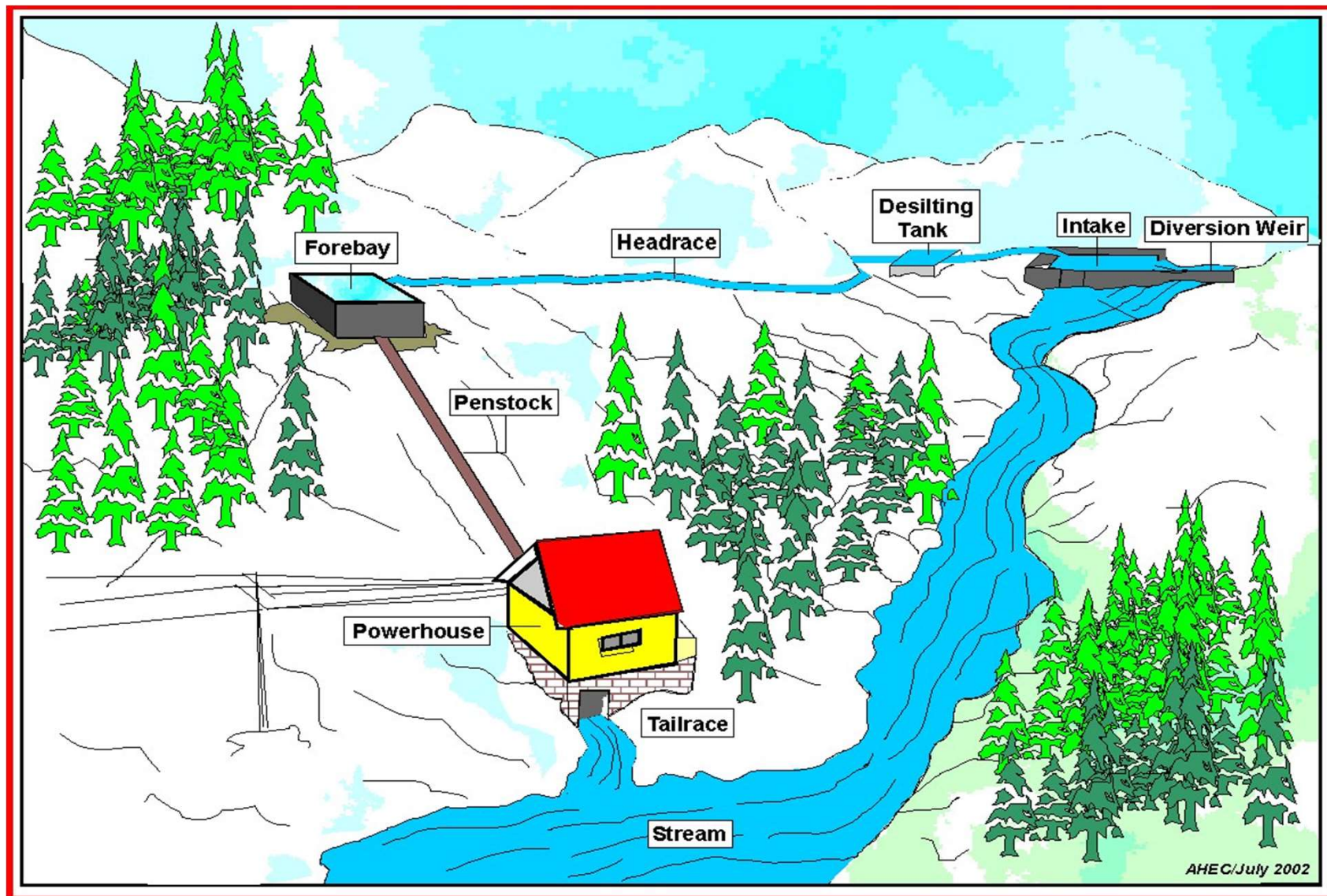
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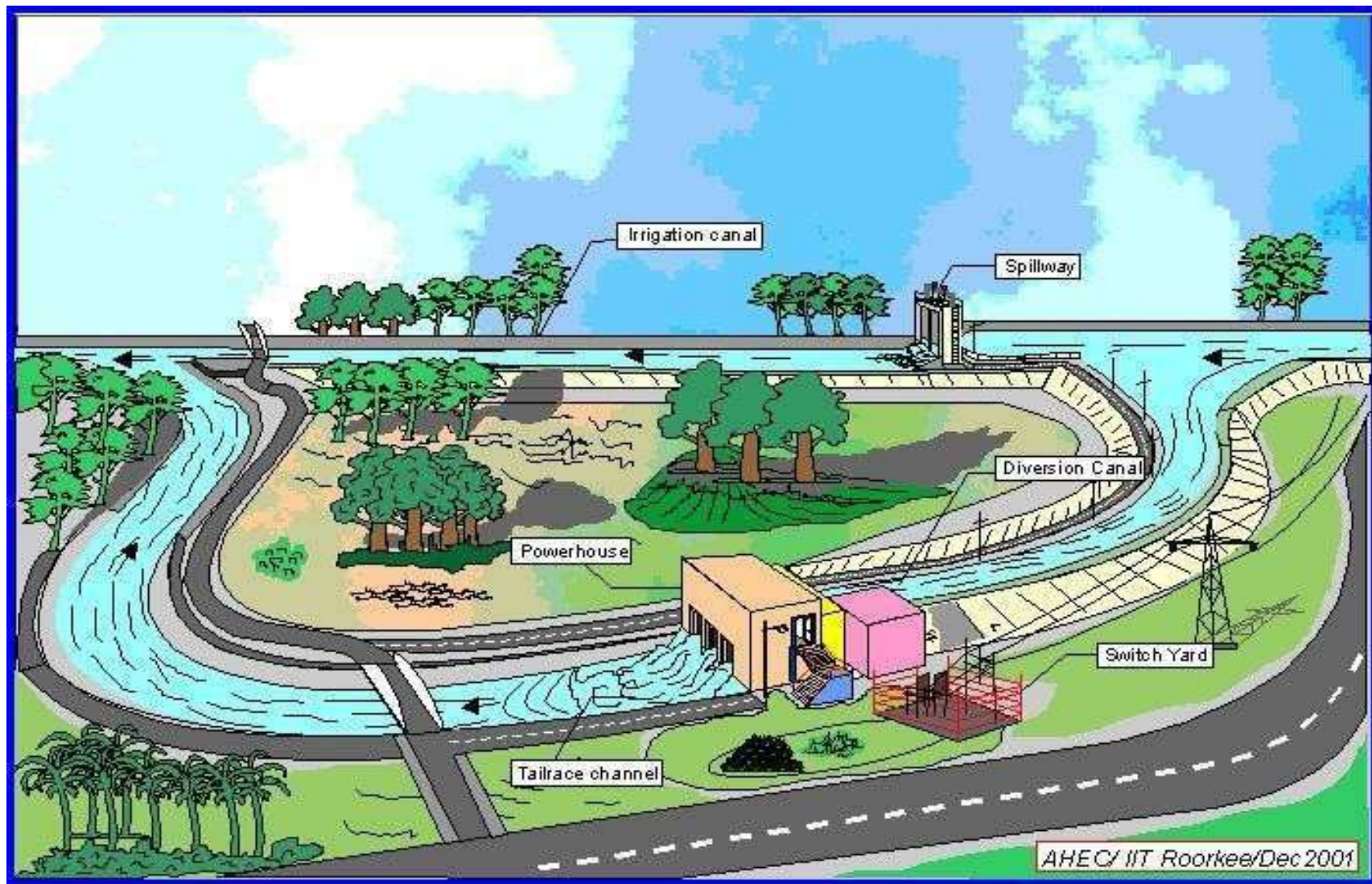


TYPE OF SHP SCHEME

- High head scheme
- Medium head scheme
- Low head scheme



Typical Arrangement of Small Hydro Power Station



Typical Arrangement of Canal Fall Small Hydro Power Station



Typical Arrangement of Dam toe Small Hydro Power Station

Calculation for Flood Discharge

EMPIRICAL METHOD

- Dicken's Formula :

Formula is especially used for hilly regions.

$$Q_{\text{flood}} = C_1 A^{3/4}$$

- Ryve's Formula

Formula is especially used for other regions.

$$Q = C_2 A^{2/3}$$

Where

C_1 = Dickens coefficient for the runoff may be taken as 14 for India.

C_2 = Ryve's coefficient for the runoff (which may be taken as 8).

Q = peak discharge in m^3/sec

A = catchment area in sq.km .

Q. Calculate the flood discharge by both methods for catchment area of 150 sqkm

Solved Example

- Calculate the flood discharge for both methods for catchment area of 150 sq km
- Dicken's Formula :

$$\begin{aligned} Q_{\text{flood}} &= C_1 A^{3/4} \\ &= 14 * (150)^{3/4} \\ &= 600 \text{ cumec} \end{aligned}$$

- Ryve's Formula

$$\begin{aligned} Q &= C_2 A^{2/3} \\ &= 8 * (150)^{2/3} \\ &= 225.848 \text{ cumec} \end{aligned}$$

DESIGN DISCHARGE

(i) Design Discharge for power generation = Q

(ii) Power channel, Forebay and spillway

Penstock and tailrace, turbine –

(10% for overloading) $(Q_p) = 1.1 * Q$

(iii) Desilting tank and intake channel (Q_d) = $(Q_p)/0.85$
(adding 15% for silt flushing)

(iv) Intake $(Q_i) = Q_d/0.90$

(adding 10% for flushing at intake)

Example

For 5 MW project having gross head of 100 m and head loss of 5%. The turbine efficiency and generator efficiency may be taken as 92% and 95% respectively. calculate design discharge of civil components

DESIGN DISCHARGE

Installed Capacity (P) = 5 MW = 5000 kW

Gross Head = 100 m

Head Losses = 5%

Net Head (H_n) = 95 m

Efficiency of turbine T_e = 92%

Efficiency of generator G_e = 95%

LeT discharge = Q

$$P = 9.81 * Q * H_n * T_e * G_e$$

$$q = 5000 / (9.81 * 0.92 * 0.95 * 95)$$

$$= 6.139 \text{ cumec}$$

DESIGN DISCHARGE FOR CIVIL STRUCTURES

(i) power channel, Forebay and spillway

$$\begin{aligned}\text{penstock and tailrace} - Q_p &= 1.1 * Q \\ &= 1.1 * 6.139 \\ &= 6.753 \text{ cumec}\end{aligned}$$

(ii) Desilting tank and intake channel (Q_d) = (Q_p)/0.85
(adding 15% for silt flushing)

$$\begin{aligned}&= 6.753/0.85 \\ &= 7.945 \text{ cumec}\end{aligned}$$

(iii) Intake (Q_i) = $Q_d/0.90$ $= 7.945/0.9$

(adding 10% for flushing at intake)

$$= 8.828 \text{ cumec}$$

DESIGN OF CIVIL WORKS COMPONENTS

- Hydraulic design
 - - to finalize the dimension of structures
- Structural design
 - - to determine the thickness and reinforcement in structures

COMPONENTS OF SMALL HYDROPOWER SCHEME

Civil Works

- Diversion and Intake structure
- Intake Channel
- Desilting tank
- Power Channel
- Forebay and spillway
- Penstock
- Surge tank
- Power house
- Tail Race
-

Alignment of Channel

- The canal length should be as short as possible.
- The canal should be aligned as high as possible usually along the contours so as to gain more head.
- To cross-drainage works such as aqueducts, flumes etc. should be avoided as far as possible.

TYPES OF CHANNEL

➤ Intake Channel, Power Channel, Spilling Channel and Tail Race Channel

- Rectangular
 - Trapezoidal
 - Open or Covered
-
- - the covered channel is rectangular in shape

Design of Channels

Area of Channel = A

Vetted Perimeter = P

Hydraulic Radius = R = A/P
Bed Slope = S

As per Manning's equation

Design Discharge (Q) of channel = $\frac{1}{n} A R^{2/3} S^{1/2}$

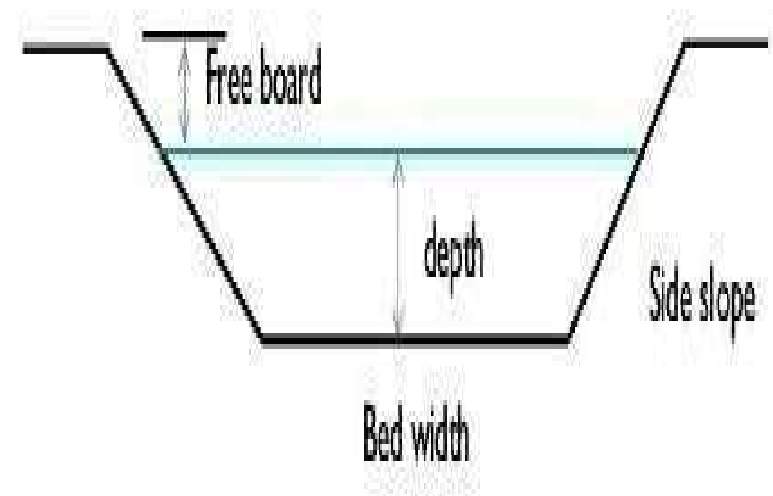
Where

n = Manning's Coefficient of roughness

(for concrete surface it can be taken as 0.018)

Free Board

- The freeboard in a power channel is determined by the following factors:
 - a) Freeboard due to surges in power channel.
 - b) Freeboard due to wave run up and effects of wind setup.
 - c) Decrease in the channel depth caused by sedimentation.
 - d) In the case of earth embankments of unlined power channel, the scour and erosion by wave action are also required to be considered.



Recommended Free Boards

Actually adopted "Free Board" for different ranges of discharge in India

Q (m ³ /s)	< 0.15	0.15 - 0.75	0.75 - 1.50	1.50 - 9.00	> 9.00
Free board (m)	0.30	0.45	0.60	0.75	0.90

Optimum Selection of Rectangular Channel

The Manning's equation provides the flow velocity in channel section. For a given section of channel, the flow velocity will be maximum if hydraulic radius is maximum that is perimeter is minimum. Let channel area, width, depth and perimeter are A, W, D, P respectively.

Consider Rectangular section

$$A = w \times d$$

$$d = A/w$$

$$P = w + 2 \times d$$

$$= w + 2A/W \quad \dots 1$$

Differentiate eq 1 with respect to w

$$dp/dw = 1 - 2A/w^2 \quad \dots 2$$

For min. or max. value of P ,

$$dp/dw = 0 \quad \dots 3$$

From eq 2 and 3

$$1 - 2A/w^2 = 0$$

$$W^2 = 2A$$

$$= 2 W * d$$

$$w = 2d$$

Double differentiate eq 2 with respect to w

$$d^2p/dw^2 = 0 + 4 A/w^3$$

$$= +ve$$

It means p is minimum

For optimize rectangular section, width is twice of depth

Design of Intake Channel

$$\text{discharge (Q)} = 7.945 \text{ cumec}$$

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} \dots\dots\dots(1)$$

$$n = 0.018$$

$$S = 1/250$$

Take rectangular optimised section

Let width of channel = w and Depth = d

Take $w = 2d$

$$\text{Area (A)} = w * d = 2d^2$$

$$\text{perimeter (P)} = w + 2 * d = 4d$$

$$\text{Hydraulic radius (R)} = A/p = 2d^2/4d = d/2$$

Trapezoidal Channel

Bed Width of channel = W

Depth of channel = d

Take $W = 2*d$

Top Width (TW) = $W + 2*d$
 $= 4d$

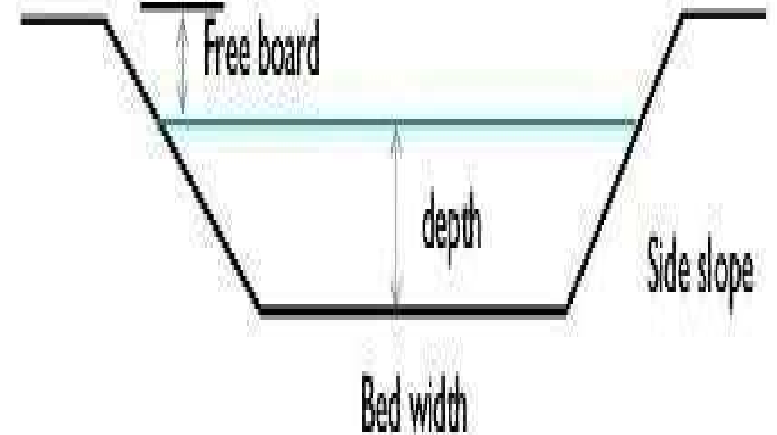
Area (A) = $(W + TW) * d / 2$
 $= (2d + 4d) * d / 2$
 $= 3d^2$

side length (sl) = $(d^2 + d^2)^{1/2}$
 $= d 2^{1/2}$

Perimeter (P) = $w + 2*sl$
 $= 2d + 2d (2)^{1/2}$
 $= 2d(1 + 2^{1/2})$

Hydraulic radius = A/P
 $= 3d^2 / 2d(1 + 2^{1/2})$
 $= 0.62 * d$

Discharging capacity = $1/n * r^{2/3} s^{1/2} a$
 $= 1/n * s^{1/2} d^{8/3} * 2.182$



Design of Intake Channel

From eqn (1)

$$7.945 = 1/0.018 * 2d^2 * (d/2)^{2/3} (1/250)^{1/2}$$

$$d^{8/3} = 7.945 * 0.018 * (250)^{1/2} / 2^{1/3}$$

$$= 1.795$$

$$d = 1.795^{3/8}$$

$$= 1.245 \text{ m}$$

$$W = 2.491 \text{ m}$$

$$\text{Free board} = 0.75 \text{ m}$$

$$\text{Total depth} = 1.245 + 0.75$$

$$= 1.995 \text{ m}$$

Design of Power Channel

$$\text{discharge (Q)} = 6.753 \text{ cumec}$$

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} \dots\dots\dots(1)$$

$$n = 0.018$$

$$S = 1/500$$

Take rectangular optimised section

Let width of channel = w and Depth = d

Take $w = 2d$

$$\text{Area (A)} = w * d = 2d^2$$

$$\text{perimeter (P)} = w + 2 * d = 4d$$

$$\text{Hydraulic radius (R)} = A/p = 2d^2/4d = d/2$$

Design of Power Channel

From eqn (1)

$$6.753 = 1/0.018 * 2d^2 * (d/2)^{2/3} (1/500)^{1/2}$$

$$d^{8/3} = 6.753 * 0.018 * (500)^{1/2} / 2^{1/3}$$

$$= 2.157$$

$$d = 2.157^{3/8}$$

$$= 1.334 \text{ m}$$

$$W = 2.668 \text{ m}$$

$$\text{Free board} = 0.75 \text{ m}$$

$$\text{Total depth} = 1.334 + 0.75$$

$$= 2.084 \text{ m}$$

FOREBAY

- A basin located just before the entrance to the penstock

PURPOSE OF FOREBAY

- To provide immediate water demand on starting the generating unit
- It can serve as a final settling basin
- The forebay provides some storage in case of sudden failure of the system
- To spill the extra water

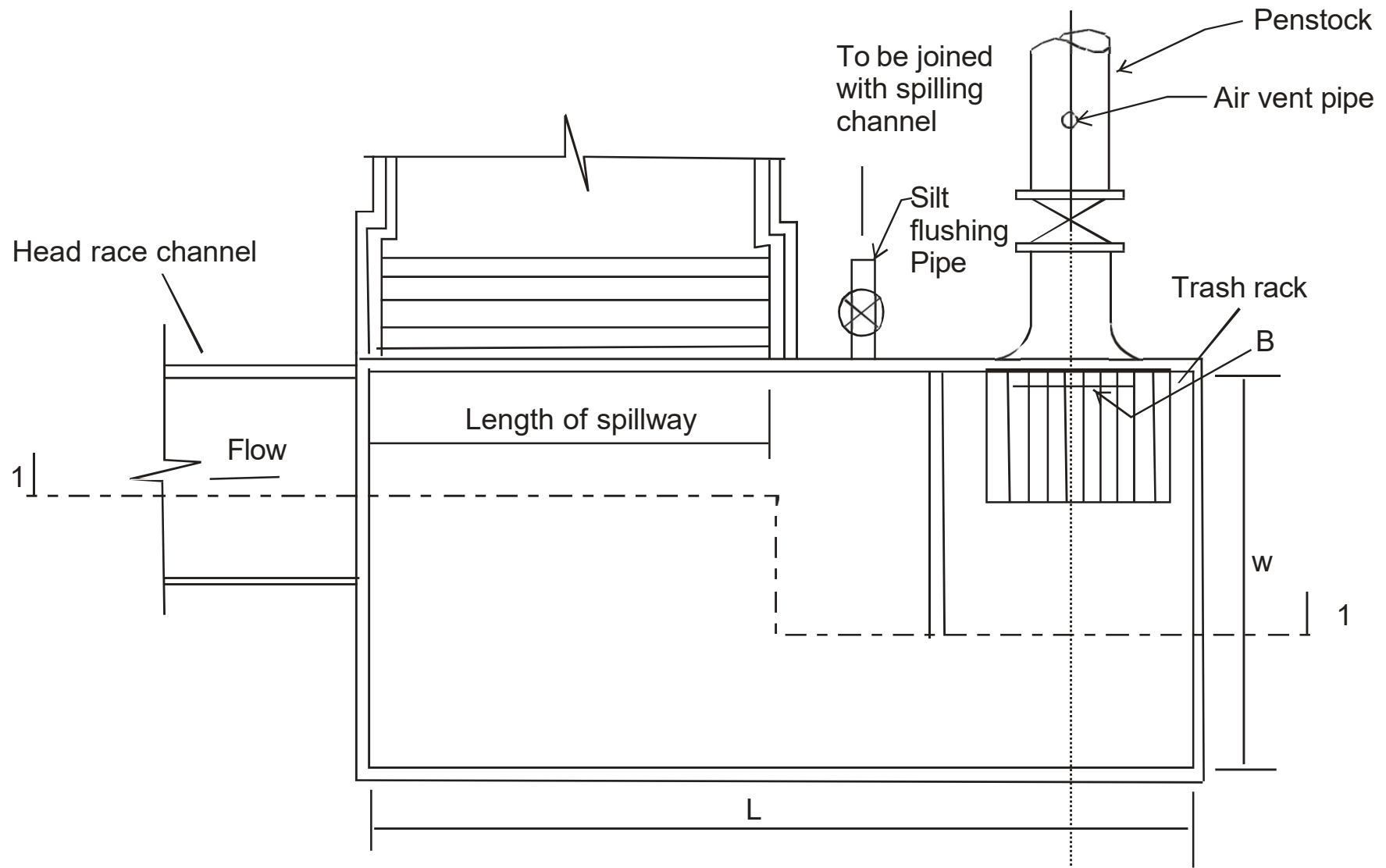
COMPONENTS OF FOREBAY

- Basin
- Spillway
- Gate or Valve
- Outlet for silt flushing
- Trashrack
- Air Vent – designed for 1 to 2% of penstock area

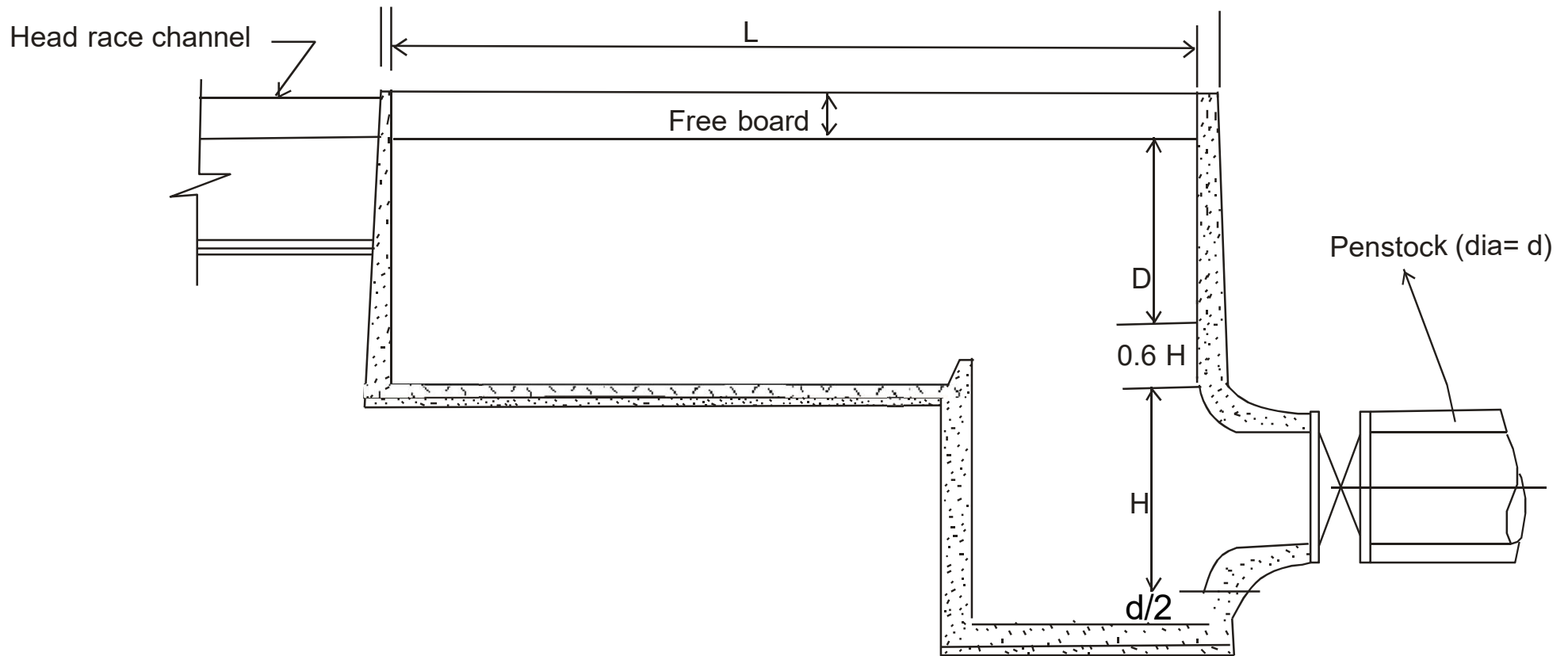
FOREBAY DESIGN

- Forebay is designed for a storage of 2 minutes
- The length of tank is kept 2 times of width and width is kept 2 times of depth. if depth of forebay tank is d then width will be $2d$ and length as $4d$
- A dead storage of about 0.5 m is provided at the bottom for silt accumulation
- Free board is provided same as adopted in intake/power channel
- Spillway is provided along one side of forebay wall
- Discharging capacity of spillway is determined from following formula
- $Q = CLH^{3/2}$
Where, C is discharge coefficient (1.7 to 2.1 adopt 1.9 m)
 L is length of crest (m)
 H is head over crest (m)

Length of crest is determined by keeping H as 0.4 m



FOREBAY



(H=1.5 *Diameter of Pipe) FOREBAY

$$\begin{aligned} \text{Additional depth} &= d/2 + 1.5*d + 0.6*1.5 d \\ &= 2.9 *d \end{aligned}$$

FOREBAY DESIGN

- Depth of forebay tank towards penstock entrance is slightly more. Depth of tank is increased due to following reasons
- (1) when the penstock is provided in penstock wall. Bell mouth is provided at the entry of penstock so as to reduce the intake losses. The bell mouth diameter is kept 1.5 times of penstock diameter. For example, if penstock diameter is 0.88 m. Then bell mouth diameter will be $1.5 * 0.88 = 1.32$ m
- (2) At the top of bell mouth, there should always be some water cushion otherwise it will suck the air. This air after reaching to turbine blades may lead to cavitation. This water cushion is also termed minimum drawdown level as that is water should not go below this level. This water cushion is kept 0.6 times of bell mouth opening that is $0.6 * 1.32 = 0.792$ m
- Total depth towards penstock side = $5.95 + 1.32 + 0.792$
- $= 8.062$ m

Length of crest is determined by keeping H as 0.4 m

FOREBAY DESIGN

- Discharging capacity of tank(Q) = 6.753 cumec,
- volume of forebay tank (V) = 6.753 x 2 x 60
- = 810.36 m³
- $V = L * W * D$
- $4 D * 2 D * D = 810.36$
- $D = (810.36/8)^{(1/3)}$
- $D = 4.662 \text{ m say } 4.7 \text{ m}$
- Depth in shallow portion (D1) = D + free board = 4.7 + 0.75 = 5.45 m
- width (w) = $2 * 4.7 = 9.4 \text{ m}$
- length (L) = $2 * 9.4 = 18.80 \text{ m}$
- ,
- +

FOREBAY DESIGN

- Additional depth in deeper portion (da) = silt storage (d/2) + bell mouth + minimum submergence
- $$= 0.44 + 1.32 + 0.79 = 2.55 \text{ m}$$
- Total depth in deeper portion (D2) = D1 + da = 5.45 + 2.55 = 8.0 m
- Length of Deeper Portion (l) = $3 * \text{Dia} + (n-1) * 2 * \text{dia}$
- $$= \text{Dia} (2n + 1)$$
- where n is no of penstock
- Width of Deeper portion (w) = $0.6 * l$

If no of penstocks and dia of penstock are 3 and 0.88 m respectively.
Calculate length and width of deeper portion

- $$l = 0.88 * (6+1) = 6.16 \text{ m}$$
- $$w = 0.6 * 6.16 = 3.69$$

FOREBAY DESIGN

- Depth in deeper portion (D2) = D1 + silt storage (d/2) + bell mouth + minimum submergence + free board
- , $= 4.7 + 0.44 + 1.32 + 0.79 + 0.75 = 7.92 \text{ m}$
- Discharging capacity of spillway is determined from following formula
- $Q = CLH^{3/2}$

Where,

C is discharge coefficient (1.9),

L is length of crest (m)

H is head over crest (m)

Example Determine Length of crest by keeping H as 0.4 m

$$\begin{aligned} L &= Q / C * H^{3/2} \\ &= 6.753 / (1.9 * 0.4^{3/2}) \\ &= 14.049 \text{ m} \end{aligned}$$

FOREBAY DESIGN

- Depth in deeper portion (D2) = D1 + silt storage (d/2) + bell mouth + minimum submergence + free board
- , $= 4.7 + 0.44 + 1.32 + 0.79 + 0.75 = 7.92 \text{ m}$
- Discharging capacity of spillway is determined from following formula
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Where,

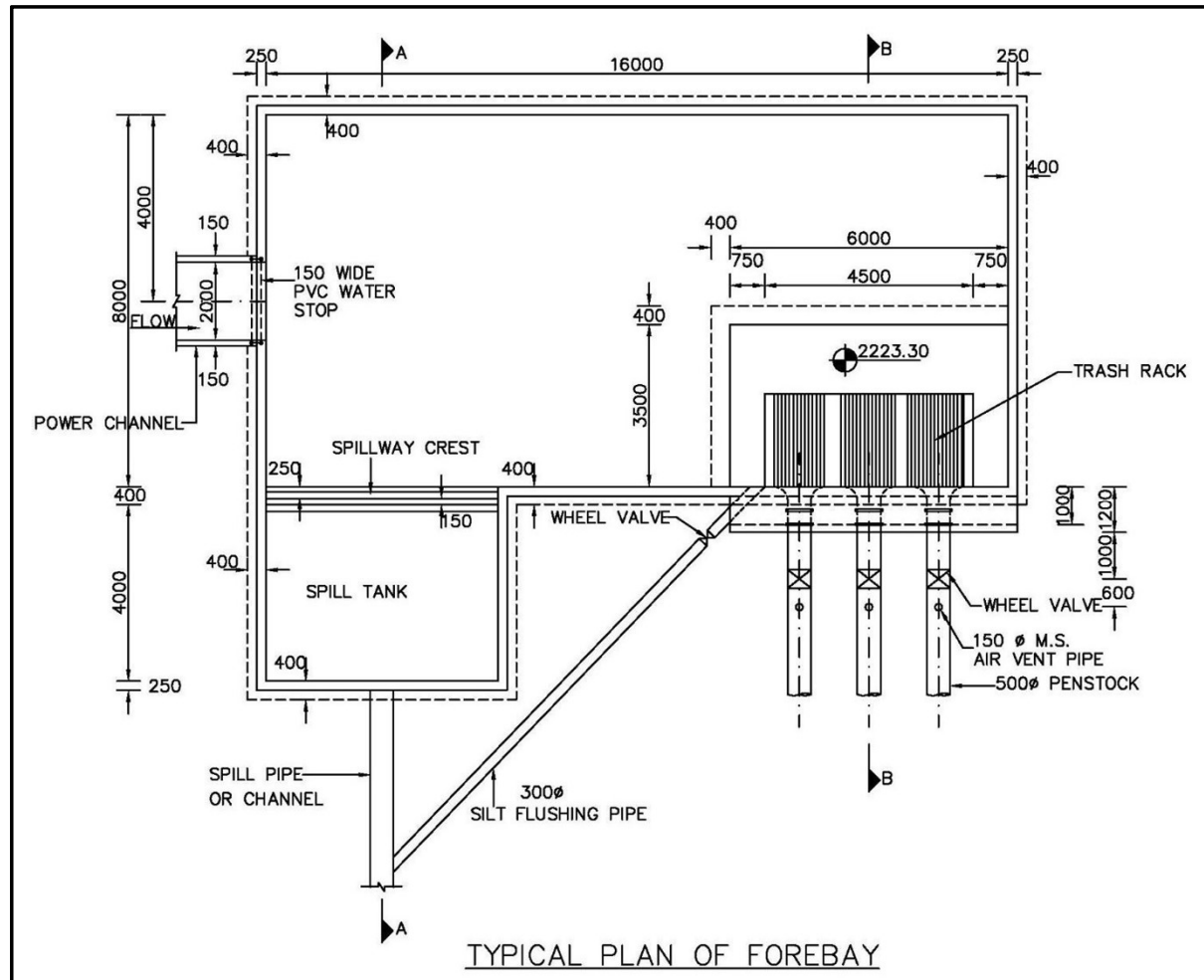
C is discharge coefficient (1.9),

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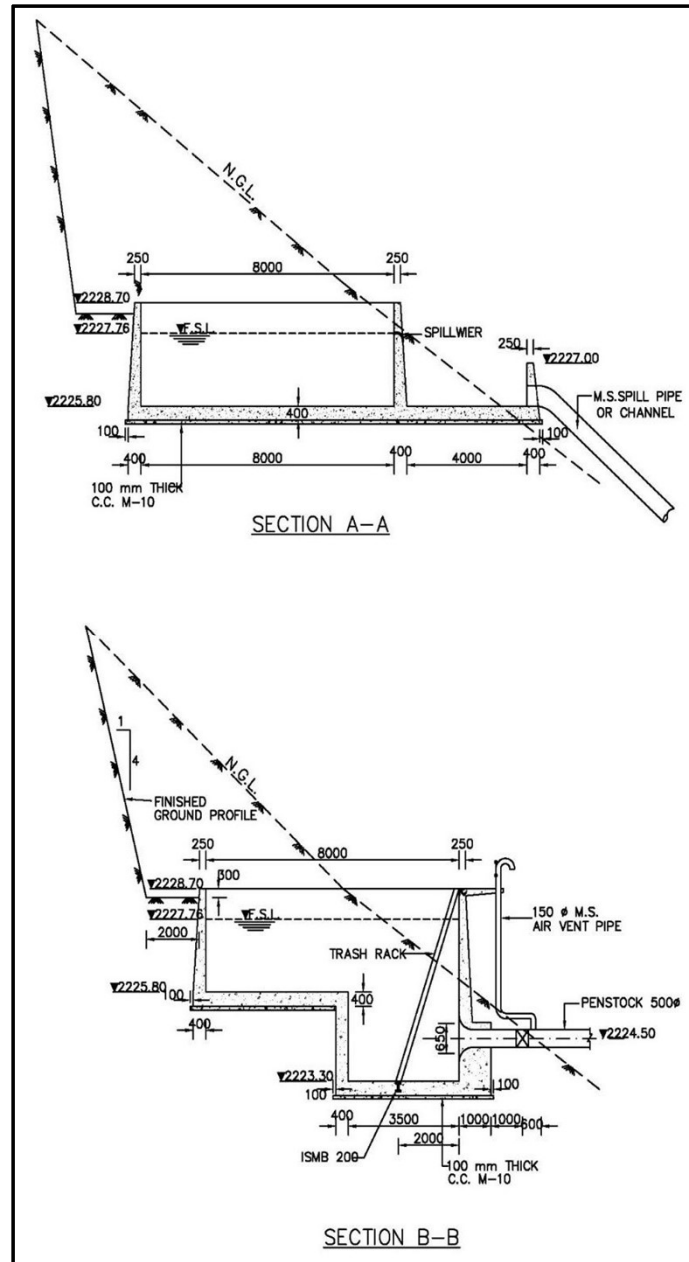
H is head over crest (m)

Example Determine Length of crest by keeping H as 0.4 m

$$\begin{aligned} L &= Q / C * H^{3/2} \\ &= 6.753 / (1.9 * 0.4^{3/2}) \\ &= 14.049 \text{ m} \end{aligned}$$



TYPICAL DETAILS OF FOREBAY (PLAN)



TYPICAL DETAILS OF FOREBAY (SECTION)

DIAMETER AND THICKNESS OF PENSTOCK

$$Q = V \times \frac{\pi D^2}{4}$$

V is velocity in m/sec.

which can be taken as 3.5

m/s. D is diameter in m

=1.56

THICKNESS OF PIPE

by stress criteria

$$\text{Thickness} = \frac{P D}{2 \sigma}, \quad \sigma \text{ is allowable stress}$$

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Example for a project having design discharge of 2.13 m. calculate diameter and thickness of pipe

$$\text{Diameter of pipe} = 2.13/3.5 = \pi D^2/4, \quad D = 0.88 \text{ m}$$

Calculation for thickness of penstock

Gross Head	=	96.33 m
Stress in steel	=	1050 kg/cm ²
Diameter of Main penstock	=	88.0 cm
Pressure rise due to water hammer	=	50% (Assured)
Total Head	=	96.33 + 48.165
	=	144.495 m
	=	144.495*0.1=14.44 kg/ cm ²

Thickness of steel liner	=	$\frac{P D}{2 \sigma}$
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Penstock	=	$\frac{14.44 \times 15}{62 \times 1050}$
	=	1.07 cm
	=	6.05 mm